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Clinical Outcomes of Transfemoral Transcatheter Mitral Valve Replacement: Results from the CHOICE-MI Registry

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Structured Abstract

Objectives

To evaluate clinical, echocardiographic, and functional outcomes following transfemoral (TF) transcatheter mitral valve replacement (TMVR) in an international registry.

Background

TMVR is an alternative for patients with mitral regurgitation (MR), but data on TF devices are limited.

Methods

The CHOICE-MI registry included patients undergoing TMVR with dedicated devices at 41 international centers. This analysis included only TF TMVR. Outcomes were assessed per MVARC criteria. Echocardiographic, functional, and clinical outcomes up to 2 years were reported. The primary endpoint was a composite of all-cause mortality or heart failure hospitalization at 2 years. Predictors were identified using multivariable Cox regression.

Results

A total of 124 patients (79 years (IQR 75–83); 50% female; EuroSCORE II 5.0% (IQR 3.3–8.5)) underwent TF TMVR with 9 different devices. Leading MR etiology was primary MR (n=49, 39.5%) with mitral annular calcification in 12.9%. Technical success was achieved in 113 of 124 patients (91.1%) and procedural mortality was low (1/124 (0.8%)). Residual MR $\leq 1+$ was achieved in 94.9% at discharge with stable results at follow-up. The rates for cardiovascular mortality were 14.7% (95% CI 7.9–21.5%) and 20.3% (95% CI 11.4–29.1%), and for the primary endpoint 44.1% (95% CI 30.6–50.0%) and 52.4% (95% CI 40.2–62.1%) at 1 and 2 years, respectively, NYHA functional class significantly improved at 1- and 2-year follow-up. Atrial fibrillation was an independent predictor of the primary endpoint, whereas technical success was associated with survival.

Conclusions

TMVR using dedicated TF devices demonstrates favorable safety, durable MR reduction, and sustained functional improvement in high-risk patients.

ClinicalTrials.gov Identifier: NCT04688190 (CHOICE-MI Registry)

Condensed Abstract

In this international, multicenter registry, 124 high-risk patients underwent transfemoral (TF) transcatheter mitral valve replacement (TMVR) using dedicated devices. Technical success was achieved in 91%, with durable MR reduction to $\leq 1+$ in $>95\%$ at 2 years. At 2 years, cardiovascular mortality was 20.3%, and the composite of all-cause mortality or heart failure hospitalization occurred in 52.4%. TF TMVR was associated with sustained clinical improvement and left ventricular reverse remodeling. Atrial fibrillation independently predicted adverse outcomes, whereas technical success conferred a survival benefit. These findings support TF TMVR as an effective and durable therapy in high-risk patients with severe MR.

Abbreviations

HF	heart failure
LVOT	left ventricular outflow tract
MAC	mitral annular calcification
MR	mitral regurgitation
MVARC	Mitral Valve Academic Research Consortium
NYHA	New York Heart Association
TEER	transcatheter edge-to-edge repair
TF	transfemoral/transseptal
TMVR	transcatheter mitral valve replacement

1 Introduction

2 Mitral regurgitation (MR) represents the most prevalent valvular heart disease in developed
3 countries, with rising incidence driven by demographic aging (1). If left untreated, chronic
4 volume overload promotes progressive heart failure (HF) and is associated with increased
5 mortality (2). Current guidelines recommend surgical mitral valve repair or replacement for
6 patients with severe symptomatic MR (3). However, up to half of affected patients are not
7 referred for surgery because of advanced age, frailty, or relevant comorbidities (4). Mitral valve
8 transcatheter edge-to-edge repair (M-TEER) has expanded treatment options for patients at high
9 surgical risk, yet anatomical limitations, residual MR, and recurrence of MR remain important
10 challenges (5,6). In this context, transcatheter mitral valve replacement (TMVR) has emerged
11 as an alternative therapeutic strategy, particularly for patients unsuitable for surgery or M-
12 TEER (7).

13 With the rapid development of dedicated TMVR systems, procedural access has become a
14 key determinant of clinical outcomes (8). Early TMVR experience largely relied on transapical
15 (TA) access, which requires a mini-thoracotomy and direct left ventricular (LV) puncture. Data
16 from transcatheter aortic valve replacement (TAVR) have reported that TA access may be
17 associated with higher procedural invasiveness, delayed recovery, and increased morbidity and
18 mortality compared with transfemoral (TF)/transseptal approaches (9). Consequently, TF
19 access has gained increasing attention as a less invasive alternative. By avoiding thoracotomy
20 and LV injury, TF TMVR has the potential to reduce periprocedural complications and improve
21 recovery. Early experiences, including valve-in-valve procedures, suggest favorable safety
22 profiles of TMVR with TF access (10).

23 Despite the growing adoption of TF TMVR, large-scale real-world data focusing specifically
24 on this access route remain limited. The present study therefore aimed to evaluate clinical

- 1 outcomes in a real-world cohort of patients undergoing TMVR with different dedicated TF
- 2 devices.

1 **Methods**

2 *Study design and study population*

3 Details of the CHOICE-MI registry (ClinicalTrials.gov Identifier: NCT04688190) have been
 4 published previously (11). Briefly, this retrospective, international, multicenter registry
 5 captured consecutive patients with symptomatic MR who were assessed for eligibility for
 6 TMVR at 41 participating centers in Europe and United Kingdom, North America, and
 7 Australia between May 2014 and December 2025. For the present analysis, only patients who
 8 ultimately underwent TF TMVR were included. Individuals who did not receive TMVR after
 9 screening, as well as those treated via TA approach, were excluded for this study (**Figure 1**).
 10 Procedural planning and decision-making were undertaken by the respective local
 11 multidisciplinary heart teams. Data on baseline characteristics and follow-up were collected at
 12 each site and subsequently pooled for central processing and statistical evaluation at the Center
 13 for Population Health Innovation (POINT) at the University Heart & Vascular Center Hamburg
 14 in Hamburg, Germany. The registry complied with the principles of the Declaration of Helsinki
 15 and received approval from the institutional review boards of the participating centres.

16 *Transcatheter mitral valve replacement devices*

17 Patients in this study were treated with nine different TMVR devices with TF approach:
 18 Altavalue (4C Medical Technologies Inc., Maple Grove, MN, USA), CardiAQ (Edwards
 19 Lifesciences, Irvine, CA, USA), Cardiovalve (Cardiovalve Ltd., Or Yehuda, Israel), Cephea
 20 (Abbott Structural, Santa Clara, CA, USA), Evoque (Edwards Lifesciences, Irvine, CA, USA),
 21 HighLife (HighLife SAS, Paris, France), Innovalve (Innovalve, Givat Shmuel, Israel), Intrepid
 22 (Medtronic Inc., Redwood City, CA, USA), and SAPIEN M3 (Edwards Lifesciences, Irvine,
 23 CA, USA). Procedural steps and implantation techniques for each device have been described
 24 previously. Postprocedural antithrombotic therapy was administered according to local

institutional protocols. Most patients received oral anticoagulation with vitamin K antagonists for at least 3–6 months following TMVR.

Echocardiographic assessment

At each participating center, MR was evaluated in accordance with current guideline recommendations (3). Etiology was categorized as primary, secondary, or mixed primary and secondary. MR severity was assessed at baseline, at discharge, and during follow-up, classified as none/trivial, mild (1+), moderate (2+), moderate-to-severe (3+), or severe (4+). Quantitative assessment of MR included measurement of the effective regurgitant orifice area (EROA), regurgitant volume, and the mean mitral transvalvular pressure gradient (MPG). Comprehensive echocardiographic evaluation further comprised stroke volume index (SVI), and LV function and dimensions, including ejection fraction (LVEF), end-diastolic diameter (LVEDD) and volume (LVEDV), as well as end-systolic diameter (LVESD) and volume (LVESV). Assessment of right ventricular (RV) function included tricuspid annular plane systolic excursion (TAPSE), pulmonary artery systolic pressure (PASP), and grading of tricuspid regurgitation (TR).

Computed tomography assessment

As part of the preprocedural evaluation, all patients underwent multislice computed tomography (CT). CT analysis included detailed assessment of mitral annular geometry, comprising inter-commissural (CC) and antero-posterior (AP) diameters, as well as annular perimeter and area. In addition, aorto-mitral angulation and the extent of mitral annular calcification (MAC) were evaluated. The severity of MAC was classified as mild, moderate, or severe according to the criteria described by Eleid et al. (12,13).

Study endpoints

The primary endpoint of the study was the composite of all-cause mortality or HF hospitalization at 2 years. Secondary 2-year endpoints comprised all-cause mortality,

cardiovascular (CV) mortality and HF hospitalizations. Echocardiographic endpoints were assessed at discharge, as well as 1- and 2-year follow-up and included residual MR, MPG, LVEF, LVEDD, LVESD, LVEDV, LVESV, TAPSE, and PASP. Functional capacity was evaluated at 1- and 2-year follow-up using the New York Heart Association (NYHA) functional class classification. Technical success, device success, and rates of in-hospital and 30-day complications were defined and reported in accordance with the criteria of the Mitral Valve Academic Research Consortium (MVARC) (14). Procedural mortality was defined as death occurring in the operating room.

Statistical analysis

Categorical variables are presented as frequency with percentage and continuous as median with interquartile range (IQR). Categorical variables are presented as counts and percentages. For proportions, 95% confidence intervals were calculated using exact binomial methods. For Kaplan–Meier analyses, 95% confidence intervals were derived from the standard error of the survival estimates. For competing-risk analyses, 95% confidence intervals were calculated based on the variance of the cumulative incidence function. Development of echocardiographic and functional outcomes over time was analyzed using mixed models with patient ID as random intercept to correctly handle intra-patient variability and utilize every information available while retaining maximum power. All mixed-effects models were specified as random-intercept models. Ordinal variables, such as MR and NYHA, were evaluated using cumulative link (CLMM) and the remaining continuous variables using linear mixed models (LMM). The proportional-odds assumption was assessed using a nominal effects test in the corresponding cumulative link models and was not violated (all $p > 0.05$). Inspection of residual plots for LMMs indicated no non-linearity, no heteroscedasticity, normality of residuals and QQ-plots of random effects verified their normality as well. However, residual-leverage plots indicated influential observations in all models, identified by high leverage, large standardized residuals,

and values beyond Cook's distance contours, therefore, we computed robust LMMs, which reduce the influence of outliers by downweighting their contribution to the model. We calculated estimated marginal means (EMMs) with 95% confidence intervals, on the mean class for CLMMs and response scale for LMMs, directly from the models and used them to plot development over time as well as to conduct Dunnett-adjusted planned contrasts for all later time points against baseline. Robust LMMs were fitted using robust scoring equations. This approach applies Huber-type down-weighting to reduce the influence of outlying residual and random-effect components and can therefore be regarded as an M-estimation-type robust mixed-model approach. To account for competing risks, Fine-Gray competing-risk analysis was used for the CV death endpoint, with non-CV death treated as the competing event. No competing-risk model was applied to the combined endpoint, as it comprised HF rehospitalization and all-cause death.

Kaplan-Meier curves for clinical endpoints up to 2-year follow-up were created accompanied by estimated mortality rates with 95% confidence intervals (95% CI). For the composite endpoint we identified relevant predictors by univariable Cox regression and included them in a multivariable model if they had $p < 0.1$ and not more than 20% missing values. The proportional hazards assumption was evaluated using a global test based on Schoenfeld residuals and was met in all models. All analyses were conducted with R version 4.5.2 (R Core Team, 2025).

1 Results

2 *Baseline characteristics*

3 A total of 124 patients underwent TF TMVR using nine dedicated devices between October
4 2016 and December 2025. A detailed device-specific distribution is provided in
5 **Supplementary Table 1**, while device-specific procedural and clinical outcomes are
6 summarized in **Supplementary Table 2**. The temporal distribution of TF TMVR procedures
7 across the study period is shown in **Supplementary Figure 1**. Baseline clinical characteristics
8 of the overall TMVR population are presented in **Table 1**. The TF TMVR cohort had a median
9 age of 79.0 years (IQR 75.0–83.0), 50% of patients were female (n=62), and the median
10 EuroSCORE II was 5.0% (IQR 3.3–8.5). The population was characterized by a substantial
11 burden of comorbidities, including atrial fibrillation (n=94, 79.7%) and coronary artery disease
12 (n=79, 66.9%), as well as relevant non-cardiac conditions such as renal dysfunction. In addition,
13 42 patients (33.9%) had a history of prior cardiac surgery. Most patients were highly
14 symptomatic, with 82.3% (n=102) classified as NYHA functional class III or IV.

15 Baseline echocardiographic and computed tomography findings are summarized in **Table 2**.
16 MR etiology was primary in 39.5% (n=49), secondary in 38.7% (n=48), and mixed primary and
17 secondary in 21.8% (n=27) of patients. Device-specific clinical outcomes according to MR
18 etiology are summarized in **Supplementary Table 3**. Median LVEF was 50% (IQR 40–58).
19 Right ventricular function showed a median TAPSE of 18mm (IQR 15–21), while pulmonary
20 pressures were elevated, with a median PASP of 48mmHg (IQR 38–59). Among patients with
21 available CT imaging, moderate or severe mitral annular calcification was present in 12.9%
22 (n=16/124).

23 *Procedural and MVARC 30-day outcomes*

Periprocedural complications and 30-day outcomes according to MVARC criteria are illustrated in **Figure 2**. Technical success, as defined by MVARC, was achieved in 113 of 124 patients (91.1%). One patient (0.8%) died intraoperatively, and conversion to open-heart surgery was necessary in six cases (5.0%). Left ventricular outflow tract (LVOT) obstruction was observed in four patients (3.3%) (**Figure 2A**).

Among MVARC-defined 30-day endpoints, acute kidney injury (AKIN stage 2 or 3) represented the most frequent adverse event, occurring in 12 of 124 patients (9.9%) of patients. Major access-site complications were reported in 9 of 124 patients (7.3%), while major or more severe bleeding events occurred in 9 of 118 (7.6%). Rates of myocardial infarction (0.9%) and disabling stroke (2.4%) were comparatively low. Within the first 30 days following TMVR, 11 patients (8.9%) died. (**Figure 2B**).

Echocardiographic outcomes

Among patients with available echocardiographic assessment at discharge MR was eliminated (i.e., none or trace residual MR) following TMVR in 87 of 117 patients (74.4%) at discharge and reduced to MR $\leq$ 1+ in 111 of 117 (94.9 %) patients.. Rates of MR elimination and reduction to MR $\leq$ 1+ were stable at follow-up (68.8% (44/ 64 patients) and 98.4% (63/64 patients), respectively, at 1 year, and 71.4% (20/28 patients) and 96.4% (27/28 patients), respectively, at 2 years) (**Figure 3A**). A CLMM testing the development over time revealed significantly lower MR at all later time points compared to baseline (all $p < 0.001$; **2**). Median transvalvular pressure gradients increased from 2.7 mmHg (IQR 2.4–3.0) at baseline to 4.1 mmHg (IQR 3.8–4.4) at discharge and remained stable thereafter, with median gradients of 4.7 mmHg (IQR 4.3–5.1) at 1 year and 4.7 mmHg (IQR 4.1–5.3) at 2-year follow-up.

EMMs and Tukey-adjusted planned contrasts of echocardiographic parameters for patients with available 1- and 2-year follow-up after TMVR are presented in **Table 3**. LVEF decreased early after TMVR, from 48.4% (95% CI 46.2–50.7) at baseline to 44.8% (95% CI 42.5–47.1)

at discharge ($p < 0.001$). This reduction persisted at 1 year (44.6% [95% CI 42.0–47.2]; $p < 0.001$ vs. baseline). At 2 years, LVEF was 46.3% (95% CI 43.1–49.5) and no longer significantly different from baseline ($p = 0.264$). In contrast, LV dimensions demonstrated progressive reverse remodeling. LVEDD decreased from 55.5 mm (95% CI 54.1–57.0) at baseline to 53.0 mm (95% CI 51.4–54.6) at discharge ($p < 0.001$). At 1 year, LVEDD measured 53.6 mm (95% CI 51.8–55.4) ($p = 0.052$ vs. baseline), and at 2 years 51.4 mm (95% CI 49.0–53.9), reflecting a significant reduction compared with baseline ($p = 0.001$). LVEDV decreased from 129.8 mL (95% CI 121.2–138.4) at baseline to 116.9 mL (95% CI 107.8–126.0) at discharge ($p = 0.008$). At 1 year, LVEDV measured 119.5 mL (95% CI 109.1–129.8) ($p = 0.098$ vs. baseline), and at 2 years 108.7 mL (95% CI 94.6–122.9), representing a significant reduction compared with baseline ($p = 0.007$). PASP remained unchanged at discharge but was significantly lower at 1 year follow-up (43.0 mmHg [95% CI 39.6–46.5]; $p = 0.025$). Echocardiographic trends for LVEDD, LVEF and PASP are shown in **Figure 4**.

Functional outcomes

Functional status according to NYHA class is shown in **Figure 3B**. Among surviving patients with available follow-up data, NYHA functional class improved compared with baseline. At baseline ($n = 124$), most patients were highly symptomatic, with 69.4% in NYHA class III and 12.9% in class IV, corresponding to 82.3% in class III/IV overall. At 1-year follow-up ($n = 63$), clinical status improved markedly, with 20.6% and 54.0% of patients in NYHA class I and II, respectively. This favorable functional profile persisted at 2 years ($n = 35$), where 14.3% of patients were in class I and 51.4% in class II, corresponding to 65.7% in NYHA class I/II. A CLMM testing the development over time revealed significantly lower NYHA at all later time points compared to baseline (all $p < 0.001$; **Supplementary Table 5**).

Clinical endpoints

The median follow-up time for the primary endpoint was 1.45 years (95% CI 0.94-2.70). Within two years, the primary endpoint occurred in 60 patients (48.4%), all-cause mortality in 39 patients (31.5%), cardiovascular mortality in 23 patients (18.9%), and HF hospitalization in 36 patients (29.8%). **Figure 5** displays the 2-year Kaplan–Meier curves for the primary endpoint of the composite of all-cause mortality or HF hospitalization, as well as for the secondary endpoints of CV mortality and all-cause mortality. The Kaplan-Meier estimated event rate for the primary composite endpoint was 44.1% (95% CI 30.6-50.0%) at 1 year and 52.4% (95% CI 40.2-62.1%) at 2 years (**Figure 5A**). The rates of all-cause mortality after TMVR were 21.2% (95% CI 13.0-28.7%) at 1 year and 31.7% (95% CI 20.5-41.4%) at 2 years (**Figure 5B**). Cardiovascular mortality accounted for 14.7% (95% CI 7.9-21.5%) at 1 year and 20.3% (95% CI 11.4-29.1%) at 2 years (**Figure 5C**).

Device-related events

During follow-up, TMVR device thrombosis was observed in 11 of 117 patients, occurring after a median of 0.82 years (IQR 0.192–1.969), while infective endocarditis occurred in 3 of 118 patients after a median of 0.96 years (IQR 0.203–2.163). Reintervention was required in 10 of 120 patients, with a median time to event of 0.77 years (IQR 0.160–1.974). A total of 10 patients (8.3%) underwent mitral valve reintervention. Of these, the majority were paravalvular leak (PVL) closures (transcatheter: n=4, surgical: n=1), mostly within 3 months post-procedure. Surgical mitral valve replacement was required in three patients (endocarditis [n=1], valve thrombosis [n=1], unknown cause [n=1]) between 3 and 8 months after the procedure. Transcatheter closure of an iatrogenic atrial septal defect (ASD) occurred in 1 case, and one patient underwent bail-out M-TEER two weeks after failed TMVR with device retrieval.

1 *Predictors of outcome after TF TMVR*

2 Univariable Cox regression analysis for the composite endpoint of all-cause mortality or HF
3 hospitalization at 2 years is presented in **Supplementary Table 7**. Among baseline clinical
4 variables, atrial fibrillation, higher body mass index, and lower estimated glomerular filtration
5 rate were associated with an increased risk of the primary endpoint in univariable analysis. In
6 multivariable Cox regression analysis, atrial fibrillation remained independently associated
7 with all-cause mortality or HF hospitalization (HR 3.97, 95% CI 1.22–12.96; $p=0.022$).

8 The adjusted association between procedural and 30-day MVARC-defined complications
9 and 2-year all-cause mortality is shown in **Supplementary Figure 2**. In multivariable Cox
10 regression analysis ($n=117$), technical success was independently associated with improved 2-
11 year survival (HR 0.31, 95% CI 0.10–0.95; $p=0.040$).

1 Discussion

2 In this multicenter, real-world registry evaluating TF TMVR with dedicated devices, we report
3 several principal findings (**Central Illustration**).

- 4 1. TMVR using nine different dedicated TF devices demonstrated high procedural
5 feasibility, achieving a technical success rate exceeding 90% despite advanced age and
6 substantial comorbidity burden of the study cohort.
- 7 2. The reduction in MR was profound and durable, with sustained elimination or reduction
8 to $\leq 1+$ in most patients up to 2 years. MR elimination translated into progressive
9 echocardiographic reverse remodeling, characterized by significant reductions in left
10 ventricular dimensions and volumes, alongside stabilization of LVEF following an early
11 decline.
- 12 3. Two-year clinical outcomes reflect the advanced disease stage of this high-risk
13 population. Technical success conferred a significant survival benefit, whereas baseline
14 atrial fibrillation emerged as a strong independent predictor of adverse outcomes.

15 *Clinical Outcomes of TF TMVR*

16 In our study, the estimated 1- and 2-year all-cause mortality rates were 21.2% and 31.7%,
17 respectively, and the composite of death or HF hospitalization occurred in 44.1% and 52.4% of
18 patients. These figures primarily reflect the substantial baseline comorbidities and advanced
19 structural heart disease of this high-risk cohort. However, when placed into the context of early
20 TMVR experiences, which were predominantly performed via a TA access, mortality rates
21 appear favorable. For instance, the 2-year mortality in the overall CHOICE-MI registry, which
22 consisted primarily of TA interventions, was reported at 38.1% (15). Similarly, the Tendyne
23 Expanded Clinical Study, which utilized a dedicated TA system in 191 patients, demonstrated
24 a 2-year mortality of 40.5% (16). The lower mortality observed in the TF cohort of this study

supports the hypothesis that avoiding a left ventricular apical puncture mitigates procedural trauma, thereby potentially improving mid-term survival.

This finding is consistent with emerging evidence from early feasibility studies of dedicated TF TMVR systems. The Intrepid TF early feasibility study reported encouraging 1- and 2-year mortality rates of 6.7% and 16.8%, respectively (17)(18). Similarly, the HighLife TSMVR feasibility study demonstrated a 1-year mortality rate of 17.0%, while the ENCIRCLE trial evaluating the SAPIEN M3 system reported a 1-year mortality rate of 13.9% (19,20). The slightly higher event rates observed in the present analysis likely reflect the real-world nature of our registry, which encompasses multiple dedicated TF TMVR devices and a broader, less selected patient population compared with the highly selected cohorts enrolled in early feasibility trials. In line with these studies, the present study demonstrated a high technical success rate of 91% and low rates of major periprocedural complications, including LVOT obstruction and procedural mortality. Taken together, these evolving data reflect the growing feasibility of fully percutaneous TF TMVR approaches; however, direct comparisons between TF and TA access remain challenging due to substantial differences in patient selection, device technology, and treatment period. A dedicated analysis addressing this question from our registry is currently ongoing.

Echocardiographic Findings

A central observation of this study is the durability of MR reduction with TF TMVR. MR elimination or reduction to $\leq 1+$ was achieved in nearly 90% at discharge and remained stable through 2-year follow-up. This contrasts with concerns regarding residual or recurrent MR observed after transcatheter mitral valve repair strategies and underscores the mechanistic advantage of complete mitral valve replacement (6). These mid-term results are encouraging and align with recent long-term data from early-generation TA systems, which demonstrated excellent durability with sustained MR reduction ($\leq 1+$ in 100% of surviving patients) and stable

hemodynamics up to 5 years. In that cohort, hemodynamic valve deterioration was exceptionally rare (1.4%), confirming that TMVR prostheses provide lasting function (21). Mean transvalvular pressure gradients increased following TMVR and remained stable at approximately 4–5 mmHg during follow-up. An increase in transmitral gradients after TMVR is expected due to implantation of a prosthetic valve within the native mitral annulus and has similarly been reported in prior TMVR studies (19). Importantly, despite this increase, mean gradients remained within a clinically acceptable range for most patients and did not appear to progressively rise beyond the early postprocedural phase. While longer follow-up of dedicated TF platforms is still required, our finding of stable valve performance up to 2 years suggests a similarly low early risk of structural valve deterioration, reinforcing TMVR as a definitive and durable treatment for this high-risk population. Nevertheless, the limited availability of echocardiographic follow-up data at 2 years warrants cautious interpretation of the long-term durability of MR reduction.

The structural consequences of MR elimination were reflected in progressive reverse remodeling. Similar ventricular remodeling following TMVR has been reported in prior studies demonstrating favorable changes in ventricular loading conditions after elimination of severe MR (7). Recent imaging analyses have further characterized favorable remodeling following TMVR, revealing not only reductions in cavity volumes but also complex geometric shifts (22)(23).

The early decline in LVEF observed after TMVR likely reflects afterload mismatch following correction of severe MR rather than intrinsic myocardial deterioration. This physiological phenomenon has long been recognized after surgical correction of chronic MR and reflects the sudden increase in effective forward afterload once the regurgitant pathway is eliminated. Recent hemodynamic analyses of TMVR investigating left- and right-ventricular hemodynamic responses after TMVR, have demonstrated similar early reductions in LVEF

despite improvements in forward hemodynamics (24). The subsequent stabilization of LVEF observed during longer follow-up in our cohort supports this interpretation and highlights the importance of longitudinal assessment when interpreting early post-procedural systolic changes.

Predictors of Adverse Outcome

In our study, atrial fibrillation emerged as the only independent baseline predictor of adverse outcomes. This observation is highly consistent with broader experiences in transcatheter mitral interventions. For instance, in patients undergoing M-TEER, pre-procedural atrial fibrillation has been confirmed as a strong independent predictor for long-term mortality, significantly increasing the risk of major adverse cardiovascular events (25). Similarly, prior population-based analyses have demonstrated that atrial fibrillation is associated with substantially higher rates of mortality and HF hospitalization following M-TEER interventions (26). Crucially, atrial fibrillation in the setting of severe MR should not be viewed merely as an isolated arrhythmia, but rather as a surrogate marker for advanced left atrial myopathy, cardiac damage and limited cardiac reserve. The negative impact of atrial fibrillation observed in the present TF TMVR cohort underscores that irreversible atrial and ventricular remodeling may impact long-term prognosis, even after optimal MR resolution. In contrast to the adverse prognostic impact of atrial fibrillation, technical success was independently associated with improved 2-year survival. This finding underscores the importance of achieving consistent, streamlined procedural performance with low periprocedural risk and high success rates, positioning technical success as a key modifiable determinant of clinical outcomes.

Device-Related Events and Long-Term Management

Device thrombosis and infective endocarditis were infrequent in the present study, with only a small number of cases observed during follow-up. Despite an overall reintervention rate of 8.3%, most reinterventions were related to PVL closure or closure of iatrogenic ASD, whereas

only a minority of cases required surgical mitral valve replacement due to endocarditis or valve thrombosis. Nevertheless, device-related thrombosis remains a clinically relevant concern in TMVR and has been increasingly recognized in the contemporary literature. In comparison, the rates of clinically significant valve thrombosis in the Intrepid TF early feasibility trial and the ENCIRCLE trial were 3.4% and 6.7%, respectively(17,27). Recent meta-analyses indicate that bioprosthetic valve thrombosis occurs more frequently in transcatheter mitral valves compared to surgical prostheses (28). These findings highlight the need for careful, patient-tailored antithrombotic strategies and warrant longer-term follow-up of patients undergoing TMVR with dedicated devices. Furthermore, continued refinement of device design aimed at improving hemodynamics and reducing thrombogenicity alongside optimized patient selection will be essential to further minimize these risks.

An important observation of the present study is the difference between technical success (91.1%) and 30-day device success (73.7%) according to MVARC criteria. While procedural feasibility was high, the lower device success rate highlights the continued occurrence of clinically relevant postprocedural complications and residual valve-related issues during the early evolution of TF TMVR technology.

Clinical Implications

Taken together, our findings demonstrate that TF TMVR has evolved into a feasible and effective therapeutic option. By avoiding myocardial injury associated with TA access, the TF approach significantly mitigates periprocedural trauma while achieving complete and durable MR elimination. The lower mortality observed in the present TF cohort compared with TA series should be interpreted cautiously, as differences in patient selection, anatomical suitability, temporal evolution of TMVR technology, and increasing operator experience likely contributed to these observations. This evolution represents a crucial step forward, offering a definitive valve replacement strategy to a frail, comorbid patient population previously deemed

unsuitable for more invasive interventions. Early identification of high-risk phenotypes will be important to tailor post-procedural surveillance, optimize antithrombotic strategies, and translate procedural success into sustained long-term survival.

Limitations

This study has several limitations that should be acknowledged. First, the analysis is based on a retrospective registry and can therefore only provide hypothesis-generating evidence. Second, the CHOICE-MI registry was designed to include multiple dedicated TMVR systems, resulting in device heterogeneity that limits the ability to draw definitive device-specific conclusions regarding procedural performance or durability. Furthermore, the inclusion of devices at different stages of clinical development – including some that are no longer available – may have introduced bias and may not accurately reflect the outcomes of currently used devices. However, the primary aim of this study was to provide a device-agnostic overview of TF TMVR outcomes across a range of systems, thereby reflecting recent clinical reality. Third, comparisons with TA TMVR cohorts are indirect and should be interpreted as descriptive only. Fourth, echocardiographic assessment was not core-lab adjudicated and relied on values provided by the participating centers. Fifth, echocardiographic and functional follow-up was incomplete, with only a minority of patients having evaluable data available at 2 years, thus limiting definitive conclusions regarding long-term durability outcomes and potentially introducing survivor bias. Sixth, assessment of functional status was limited to physician-reported NYHA functional class, while standardized patient-reported outcome measures and quality-of-life assessments were not systematically available. Additionally, analyses of NYHA functional class at later follow-up time points were limited to surviving patients with available follow-up data and may therefore be affected by substantial survivor bias. Seventh, detailed baseline and follow-up antithrombotic medication data were not systematically available across participating centers, limiting interpretation of thrombotic outcomes after TMVR. Last, longer-

- 1 term follow-up is required to assess the incidence of structural valve deterioration and late
- 2 device failure as well as the prognostic impact of device thrombosis

Conclusion

In this large, real-world cohort, TF TMVR using different dedicated devices was associated with high technical success, durable elimination of MR, functional improvement, and echocardiographic reverse remodeling in patients with available follow-up data. Atrial fibrillation and technical success emerged as key baseline and procedural determinants of outcomes, underscoring the importance of timely intervention and optimal procedural planning. Collectively, these findings support TF TMVR as an effective and durable therapeutic option and reinforce its continued evolution and integration into the contemporary management of mitral valve disease.

Perspectives

What is known?

Transfemoral transcatheter mitral valve replacement is emerging as a promising alternative therapy for patients with severe mitral regurgitation who are unsuitable for surgery or transcatheter edge-to-edge repair.

What is new?

These real-world data from a contemporary registry including multiple devices demonstrate that TF TMVR represents a feasible therapy offering high procedural success, durable elimination of mitral regurgitation and meaningful clinical improvement in a high-risk population.

What is next?

Future studies should focus on refining patient selection, optimizing antithrombotic strategies, and establishing long-term durability, while randomized comparisons with alternative therapies will be critical to define the role of TF TMVR within contemporary treatment algorithms.

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Figure Legends

Figure 1. Title: Study flow chart

Legend: The CHOICE-MI registry included consecutive patients undergoing TMVR at 41 international centers. This study included only patients undergoing TMVR with dedicated transfemoral devices (N=124). Patients with transapical access (N=421) were excluded from this analysis.

Abbreviations:

MR: mitral regurgitation, TA: transapical, TF: transfemoral, TMVR: transcatheter mitral valve replacement

Figure 2. Title: Procedural and 30-day MVARC outcomes

Legend: Procedural outcomes after TF TMVR (2A) and 30-day MVARC outcomes (2B) are summarized in bar plots. Each bar depicts the respective event rate (%). Overall complication rates were low with a high technical success rate.

Abbreviations:

AKIN: Acute Kidney Injury Network, LVOT: left ventricular outflow tract, MVARC: Mitral Valve Academic Research Consortium

Figure 3. Title: Mitral regurgitation and functional outcomes

Legend: Mitral regurgitation (MR) (3A) and New York Heart Association (NYHA) functional class (3B) at baseline and follow-up are presented in bar plots. Compared to baseline, both MR (discharge, 1- and 2-year follow-up) and NYHA functional class (1- and 2-year follow-up) improved significantly.

* $p < 0.001$ for the comparison between baseline vs. discharge (only for MR), baseline vs. 1-year follow-up (MR and NYHA), and baseline vs. 2-year follow-up (MR and NYHA)

Abbreviations:

MR: mitral regurgitation, NYHA: New York Heart Association

Figure 4. Title: Echocardiographic trends for LVEDD, LVEF and PASP

Legend: Time course of left ventricular ejection fraction (A), left ventricular end-diastolic diameter (B), and pulmonary artery systolic pressure (C) at baseline, discharge, 1-year, and 2-year follow-up. Values are shown as means with error bars (e.g., $\pm$ standard deviation/standard error), with corresponding sample sizes (n).

Abbreviations:

LVEDD: left ventricular end-diastolic diameter, LVEF: left ventricular ejection fraction, PASP: pulmonary artery systolic pressure

Figure 5. Title: Two-year clinical endpoints

Legend: For Kaplan–Meier analyses, 95% confidence intervals were derived from the standard error of the survival estimates. For competing-risk analyses, 95% confidence intervals were calculated based on the variance of the cumulative incidence function. These are shown for the primary endpoint (A), all-cause mortality (B), and the cardiovascular mortality (C).

Abbreviations:

HF: heart failure

Central Illustration. Title: Clinical outcomes of transfemoral transcatheter mitral valve replacement in the cCHOICE-MI registry

Legend: In this international multicenter registry, 124 high-risk patients underwent TF TMVR using nine dedicated devices between 2016 and 2025. Technical success was achieved in 91.1% with low procedural mortality (0.8%). Among patients with available follow-up, MR reduction to $\leq 1+$ was achieved in approximately 95% at discharge and remained sustained through 2 years. TF TMVR was associated with favorable echocardiographic remodeling and symptomatic improvement during follow-up. Technical success was associated with improved survival, whereas atrial fibrillation independently predicted adverse clinical outcomes.

Abbreviations:

CHOICE-MI: Choice of Optimal Transcatheter Treatment for Mitral Insufficiency, HFH: heart failure hospitalization, TF: transfemoral, TMVR: transcatheter mitral valve replacement, UK: United Kingdom

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Tables

Table 1: Clinical baseline characteristics

Parameter	TF TMVR (N=124)
Clinical Baseline Parameters	
Age (years)	79.0 (75.0, 83.0)
Female sex, n (%)	62 (50.0)
BMI (kg/m ²)	25.5 (23.5, 28.3)
EuroSCORE II (%)	5.0 (3.8, 8.5)
STS PROM (MV replacement) (%)	6.2 (4.3, 8.7)
STS PROM (MV repair) (%)	3.7 (2.4, 5.8)
Atrial fibrillation, n (%)	94 (79.7)
Hypertension, n (%)	104 (84.6)
Diabetes, n (%)	38 (30.6)
Extracardiac arteriopathy, n (%)	12 (11.2)
Coronary artery disease, n (%)	79 (66.9)
Prior myocardial infarction, n (%)	37 (29.8)
Prior CABG, n (%)	34 (27.4)
Prior PCI, n (%)	54 (43.5)
COPD, n (%)	26 (21.0)
Prior stroke, n (%)	8 (6.5)
eGFR (ml/min/1.73 m ²)	52.0 (35.0, 66.0)
Prior dialysis	2 (1.6)
HF hospitalization (prior 12 months), n (%)	68 (54.8)
NYHA Functional Class III/IV, n (%)	102 (82.3)
NTproBNP (pg/mL)	1402.0 (817.8, 3107.5)
Heart Failure Medication	
Betablocker, n (%)	96 (81.4)
ACE-inhibitor/ARB, n (%)	66 (55.9)
ARNI, n (%)	22 (18.6)
MRA, n (%)	45 (38.1)
SGLT-2 inhibitor, n (%)	32 (27.6)

Abbreviations:

ACE: angiotensin converting enzyme, ARB: angiotensin II receptor blocker, ARNI: angiotensin receptor-neprilysin inhibitor, BMI: body mass index, CABG: coronary artery bypass graft, COPD: chronic obstructive pulmonary disease, eGFR: estimated glomerular filtration rate, HF: heart failure,

MRA: mineralocorticoid receptor antagonists, MV: mitral valve, NTproBNP: N-terminal pro-brain natriuretic peptide, NYHA: New York Heart Association, PCI: percutaneous coronary intervention, SGLT-2: sodium-glucose cotransporter-2, STS PROM: Society of Thoracic Surgeons Predicted Risk of Mortality, TMVR: transcatheter mitral valve replacement

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Table 2: Echocardiographic and CT parameters at baseline

Parameter	TF TMVR (N=124)
Echocardiography	
Primary MR, n (%)	49 (39.5)
Secondary MR, n (%)	48 (38.7)
Mixed primary/secondary MR, n (%)	27 (21.8)
Mitral stenosis (severe), n (%)	3 (2.5)
MPG (mmHg)	2.0 (2.0, 4.0)
EROA (mm ²)	0.30 (0.2, 0.4)
RVol (mL)	51.5 (39.0, 68.8)
LAV (mL)	98.6 (68.5, 124.8)
LVEF (%)	50.0 (40.0, 57.8)
LVEDD (mm)	55.0 (50.0, 61.8)
LVEDV (mL)	121.5 (101.6, 154.5)
LVESD (mm)	40.0 (34.5, 47.0)
LVESV (mL)	63.0 (47.0, 84.0)
PASP (mmHg)	48.0 (38.0, 59.0)
TAPSE (mm)	18.0 (15.0, 21.0)
TR $\geq 3+$, n (%)	65 (52.4)
Computed Tomography	
CC diameter, mm	41.0 (38.0, 44.0)
SL diameter, mm	36.0 (31.2, 38.8)
Annulus perimeter, mm	127.4 (116.0, 136.3)
Annulus area, cm ²	12.5 (10.8, 14.0)
Aorto-mitral angulation, °	125.4 (115.9, 132.0)
MAC ($\geq$ moderate), n (%)	16 (12.9)

Abbreviations:

SL:septal-lateral, CC: intercommissural, CT: computed tomography EROA: effective regurgitant orifice area, LAV: left atrial volume, LVEDD: left ventricular end-diastolic diameter, LVEDV: left ventricular end-diastolic volume, LVEF: left ventricular ejection fraction, LVESD: left ventricular end-systolic diameter, LVESV: left ventricular end-systolic volume, MAC: mitral annulus calcification, MPG: mean transvalvular pressure gradient, MR: mitral regurgitation, PASP: pulmonary artery systolic pressure, RVol: regurgitant volume, SVI: stroke volume index, TAPSE: tricuspid annular plane systolic excursion, TMVR: transcatheter mitral valve replacement, TR: tricuspid regurgitation

Table 3: Planned contrasts of echocardiographic parameters at discharge, 1- and 2- year follow-up after TF TMVR

Baseline vs. Discharge					
Echocardiography Endpoints	n (baseline)/ n (discharge)	Baseline	Discharge	Change to Baseline	p-value
MPG, mmHg	109/107	2.7 (2.4, 3.0)	4.1 (3.8, 4.4)	1.32 (0.85, 1.80)	<0.001
LVEF, %	124/115	48.4 (46.2, 50.7)	44.8 (42.5, 47.1)	-3.6 (-5.4, -1.8)	<0.001
LVEDD, mm	118/90	55.5 (54.1, 57.00)	53.0 (51.4, 54.6)	-2.5 (-4.2, -0.9)	0.001
LVEDV, mL	76/63	129.8 (121.2, 138.4)	116.87 (107.8, 126.0)	-12.9 (-23.1, -2.8)	0.008
LVESD, mm	87/63	41.4 (39.3, 43.5)	40.15 (37.8, 42.4)	-1.3 (-3.6, 1.0)	0.409
LVESV, mL	75/64	67.1 (60.6, 73.7)	62.9 (56.2, 69.7)	-4.2 (-10.2, 1.8)	0.242
TAPSE, mm	88/63	18.0 (16.9, 19.0)	17.7 (16.6, 18.9)	-0.2 (-1.7, 1.3)	0.953
PASP, mmHg	111/81	47.8 (45.2, 50.4)	46.3 (43.3, 49.3)	-1.5 (-5.2, 2.3)	0.662

Baseline vs. 1-Year Follow-up					
Echocardiography Endpoints	n (baseline)/ n (discharge)	Baseline	1-Year Follow-up	Change to Baseline	p-value
MPG, mmHg	109/62	2.7 (2.4, 3.0)	4.7 (4.3, 5.1)	2.0 (1.4, 2.5)	<0.001
LVEF, %	124/63	48.4 (46.2, 50.7)	44.6 (42.0, 47.2)	-3.8 (-6.1, -1.6)	<0.001
LVEDD, mm	118/57	55.51 (54.1, 57.0)	53.6 (51.8, 55.4)	-2.0 (-3.9, 0.0)	0.052
LVEDV, mL	76/42	129.8 (121.2, 138.4)	119.45 (109.1, 129.8)	-10.4 (-22.1, 1.4)	0.098
LVESD, mm	87/41	41.4 (39.3, 43.5)	41.92 (39.4, 44.5)	0.5 (-2.2, 3.2)	0.923
LVESV, mL	75/39	67.1 (60.6, 73.7)	65.8 (58.3, 73.3)	-1.3 (-8.5, 5.8)	0.918

TAPSE, mm	88/43	18.0 (16.9, 19.0)	17.0 (15.7, 18.4)	-0.9 (-2.6, 0.7)	0.406
PASP, mmHg	111/57	47.8 (45.2, 50.4)	43.0 (39.6, 46.5)	-4.8 (-9.0, -0.5)	0.025
Baseline vs. 2-Year Follow-up					
Echocardiography Endpoints	n (baseline)/ n (discharge)	Baseline	2-Year Follow-up	Change to Baseline	p-value
MPG, mmHg	109/26	2.7 (2.4, 3.0)	4.7 (4.1, 5.3)	2.0 (1.2, 2.8)	<0.001
LVEF, %	124/27	48.4 (46.2, 50.7)	46.3 (43.1, 49.5)	-2.2 (-5.4, 1.0)	0.264
LVEDD, mm	118/25	55.5 (54.1, 57.0)	51.4 (49.0, 53.9)	-4.1 (-6.8, -1.3)	0.001
LVEDV, mL	76/18	129.8 (121.2, 138.4)	108.7 (94.6, 122.9)	-21.1 (-37.6, -4.6)	0.007
LVEDS, mm	87/20	41.4 (39.3, 43.5)	41.4 (39.3, 43.5)	-1.7 (-5.3, 2.0)	0.567
LVESV, mL	75/15	67.1 (60.6, 73.7)	62.5 (52.6, 72.4)	-4.6 (-15.1, 5.9)	0.584
TAPSE, mm	88/19	18.0 (16.9, 19.0)	17.4 (15.5, 19.3)	-0.6 (-2.9, 1.7)	0.854
PASP, mmHg	111/27	47.8 (45.2, 50.4)	42.5 (37.8, 47.2)	-5.3 (-11.1, 0.5)	0.085

Estimated marginal means (EMMs) with 95% confidence intervals are shown, p-values are Tukey-adjusted

Abbreviations:

LVEDD: left ventricular end-diastolic diameter, LVEDV: left ventricular end-diastolic volume, LVEF: left ventricular ejection fraction, LVESD: left ventricular end-systolic diameter, LVESV: left ventricular end-systolic volume, MR: mitral regurgitation, n/a: not available, PASP: pulmonary artery systolic pressure, TAPSE: tricuspid annular plane systolic excursion, TMVR: transcatheter mitral valve replacement, TR: tricuspid regurgitation

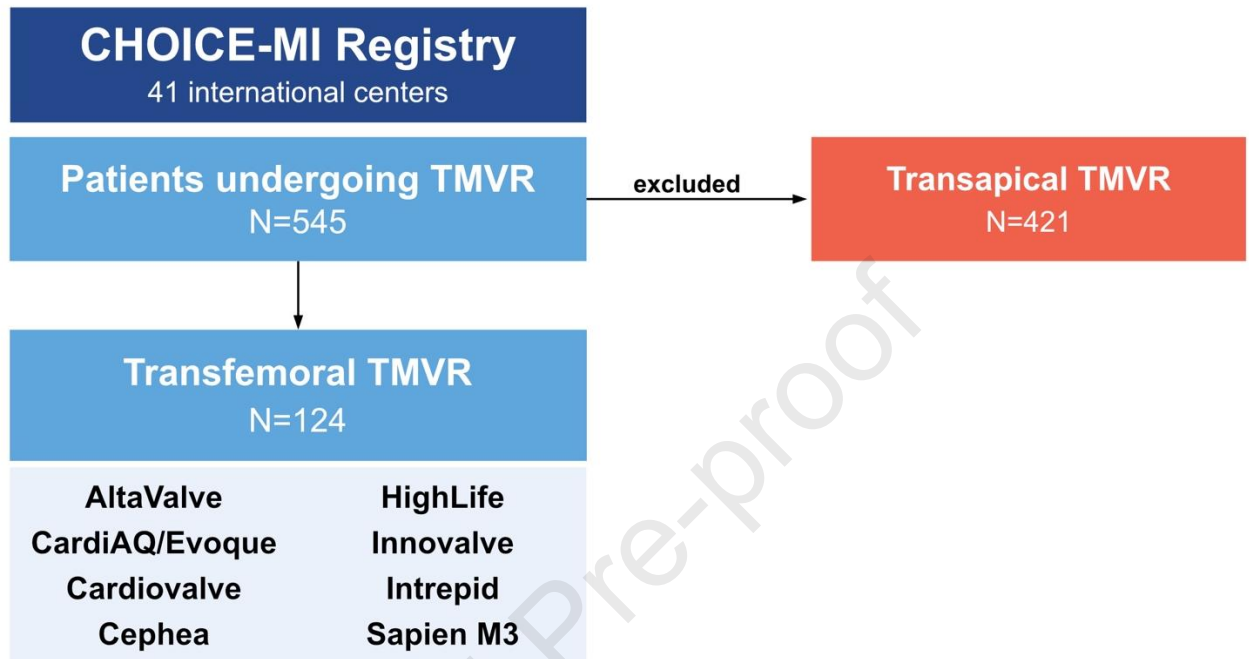
Figure 1

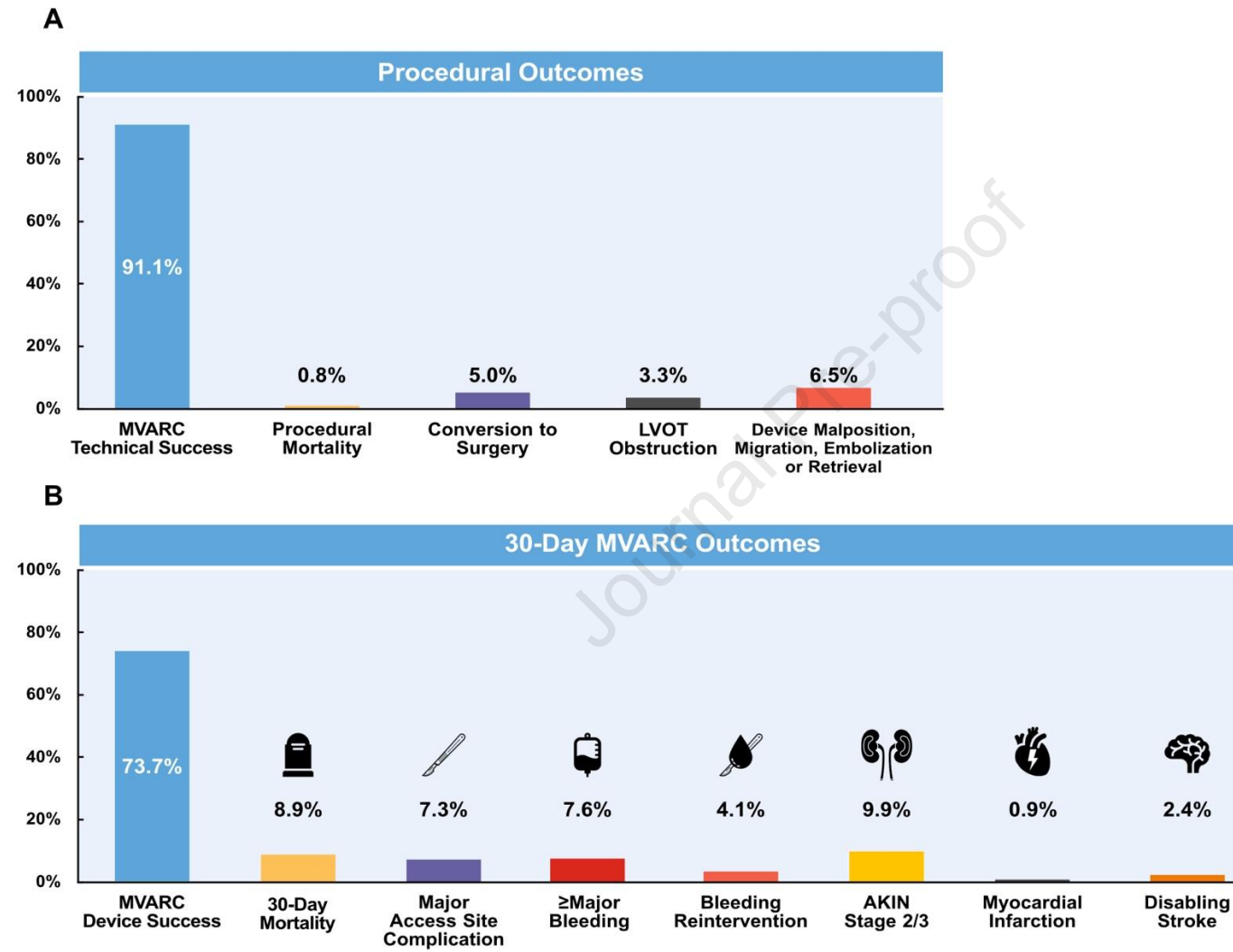
Figure 2

Figure 3

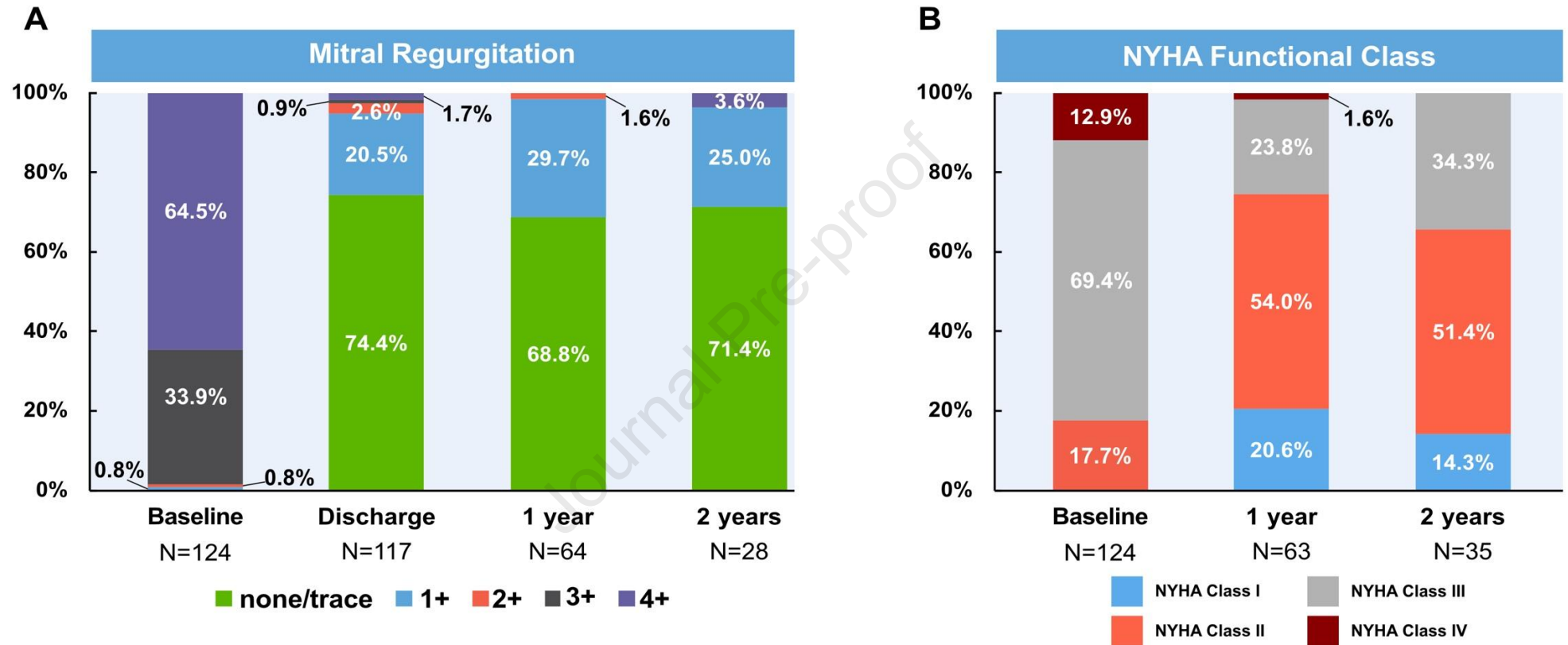


Figure 4

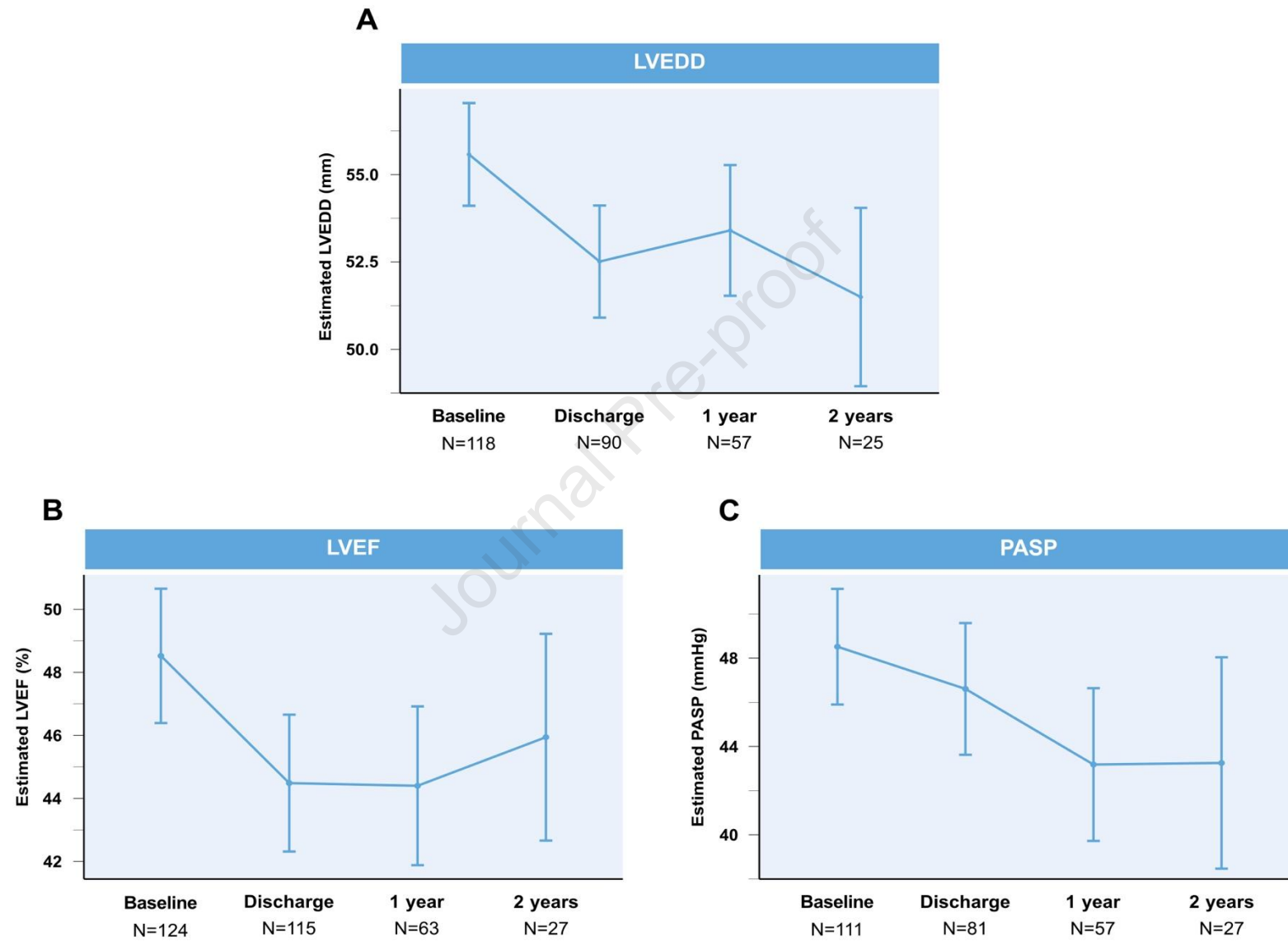
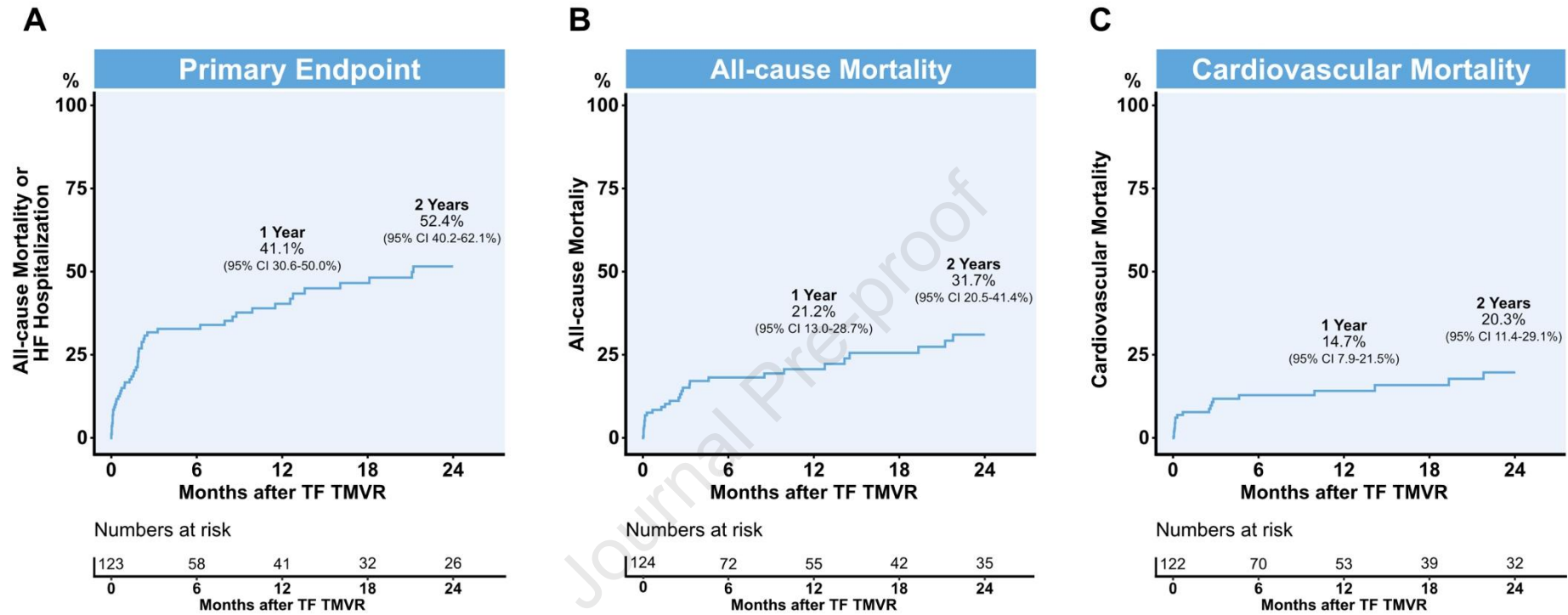
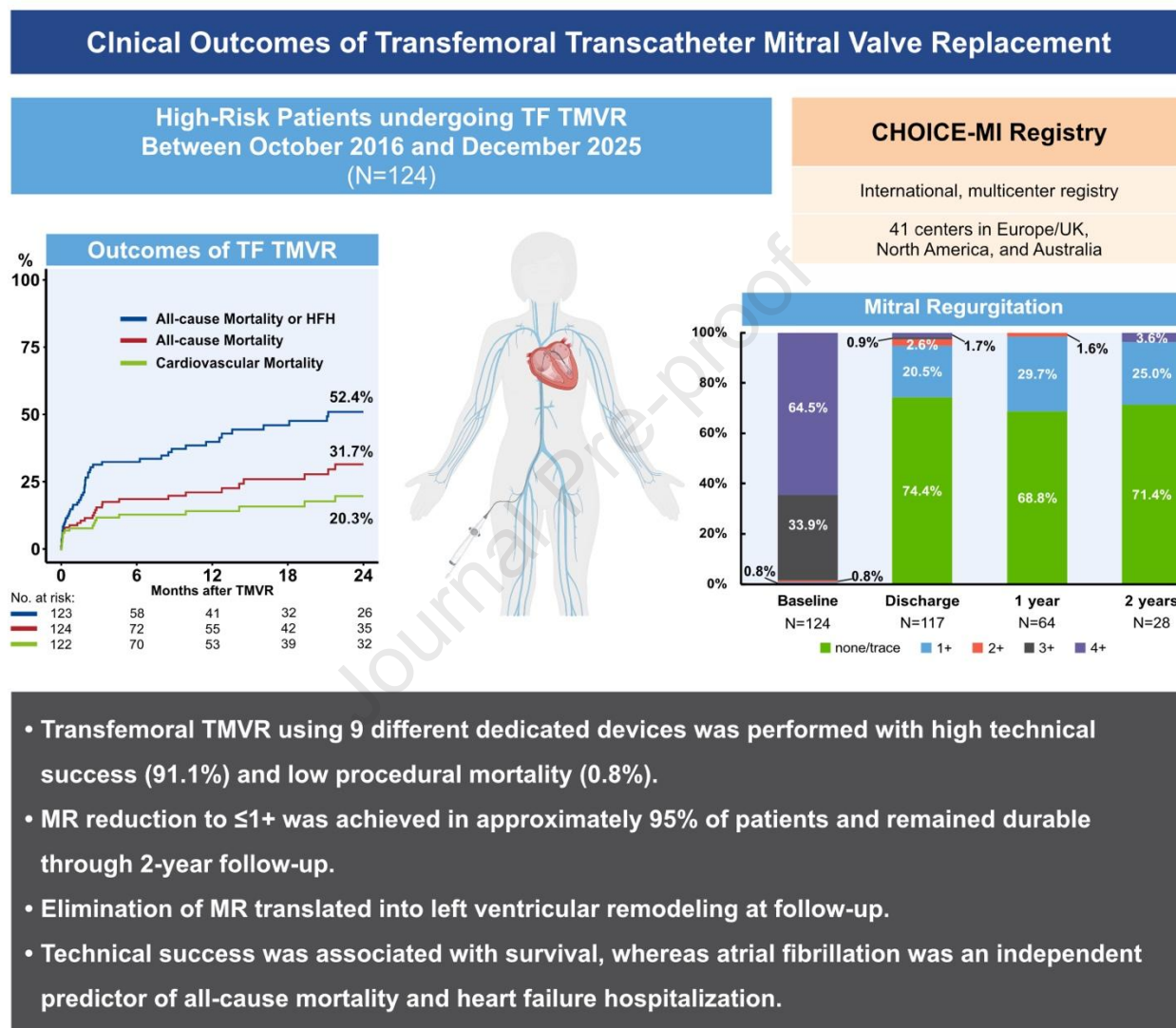


Figure 5



Central Illustration



Supplement

Supplementary Table 1: Distribution of TF TMVR devices

TF TMVR Devices	Number of Patients
	(N=124)
AltaValve (4C Medical), n (%)	18 (14.5)
CardiAQ (Edwards Lifesciences), n (%)	3 (2.4)
Cardiovalve (Venus Medtech), n (%)	8 (6.5)
Cephea (Abbott), n (%)	12 (9.7)
Evoque (Edwards Lifesciences), n (%)	7 (5.6)
HighLife (HighLife Medical), n (%)	30 (24.2)
Innovalve (Innovalve Bio Medical), n (%)	4 (3.2)
Intrepid (Medtronic), n (%)	4 (3.2)
Sapien M3 (Edwards Lifesciences), n (%)	38 (30.6)

Abbreviations:

TF: transfemoral, TMVR: Transcatheter Mitral Valve Replacement

Supplementary Table 2: Device-specific outcomes

Device	N	Technical success, n (%), (95% CI)	Primary endpoint (1 year), n (%), (95% CI)	Primary endpoint (2 years), n (%), (95% CI)
SAPIEN M3	38	36 (94.7%) 82.3-99.4%)	7 (21.2%) (5.1-34.1%)	10 (36.4%) (14.0-52.9%)
HighLife	30	27 (90.0%) (73.5-97.9%)	14 (55.8%) (29.4-72.3%)	15 (62.1%) (33.8-78.3%)
AltaValve	18	17 (94.4%) (72.7-99.9%)	10 (67.3%) (24.6-85.8%)	10 (44.4%) (30.8-78.5%) *
Cephea	12	10 (83.3%) (51.6-97.9%)	4 (33.3%) (0.5-55.3%)	5(44.4%) (5.0-67.5%)
Cardiovalve	8	7 (87.5%) (47.3-99.7%)	3 (37.5%) (8.5-75.5%) *	3 (37.5%) 8.5-75.5%) *
Evoque	7	6 (85.7%) 42.1-99.6%)	2 (28.6%) (0.0-55.3%)	4 (57.1%) (0.0-81.8%)
Innovalve	4	4 (100.0%) (39.8-100.0%)	0 (0.0%) (0.0-0.0%)	0 (0.0%) (0.0-0.0%)
Intrepid	4	4 (100.0%) (39.8-100.0%)	1 (33.3%) (0.0-70.0%)	1 (25%) (0.6-80.6%) *
CardiAQ	3	2 (66.7%) (9.4-99.2%)	3 (100.0%) (29.2-100.0%) *	3(100%) (29.2-100.0%) *

* raw event rates not estimable by Kaplan Meier estimator.

The primary endpoint was a composite of all-cause mortality or heart failure hospitalization.

Abbreviations:

CI: confidence interval; CV: cardiovascular

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Supplementary Table 3: Device specific clinical outcomes according to MR etiology

Group	Technical success n/N (%), (95% CI)	Primary endpoint (1 year) n (%), (95% CI)	Primary endpoint (2 years) n (%), (95% CI)	CV mortality (1 year) n (%), (95% CI)	CV mortality (2 years) n (%), (95% CI)
Primary MR	43/46 (93.5%) (82.1-98.6%)	19 (44.6%) (27.1-57.9%)	24 (64.2%) (42.5-77.7%)	8 (19.6%) (7.0-32.1%)	10 (22.7%) (11.5-37.8%) *
Secondary MR	40/45 (88.9%) (75.9-96.3%)	15 (38.8%) (20.4-52.9%)	16 (43.2%) (23.3-57.9%)	5 (12.3%) (2.0-22.6%)	5 (11.1%) (3.7-24.1%) *
Mixed MR	29/32 (90.6%) (75.0-98.0%)	10 (39.9%) (16.2-56.3%)	11 (46.2%) (19.8-63.9%)	3 (10.7%) (0.00-22.7%)	4 (12.5%) (3.5-29.0%) *

* raw event rates not estimable by Kaplan Meier estimator.

The primary endpoint was a composite of all-cause mortality or heart failure hospitalization.

Abbreviations

CI: confidence interval; CV: cardiovascular ; MR: mitral regurgitation

Supplementary Table 4: Availability and reasons for missing follow-up values

Parameter	Timepoint	Available, n (%)	Missing due to death, n (%)	FU not yet reached, n (%)	Missing due to lost FU, n (%)	Missing other/not documented, n (%)
MR	1 year	64 (51.6%)	20 (16.1%)	15 (12.1%)	23 (18.5%)	2 (1.6%)
MR	2 years	28 (22.6%)	29 (23.4%)	36 (29.0%)	22 (17.7%)	9 (7.3%)
NYHA	1 year	63 (50.8%)	21 (16.9%)	15 (12.1%)	23 (18.5%)	2 (1.6%)
NYHA	2 years	35 (28.2%)	28 (22.6%)	36 (29.0%)	20 (16.1%)	5 (4.0%)
LVEDD	1 year	57 (46.0%)	23 (18.5%)	16 (12.9%)	23 (18.5%)	5 (4.0%)
LVEDD	2 years	25 (20.2%)	29 (23.4%)	36 (29.0%)	22 (17.7%)	12 (9.7%)
LVEF	1 year	63 (50.8%)	21 (16.9%)	15 (12.1%)	23 (18.5%)	2 (1.6%)
LVEF	2 years	27 (21.8%)	29 (23.4%)	36 (29.0%)	22 (17.7%)	10 (8.1%)

Abbreviations:

MR: mitral regurgitation, NYHA: New York Heart Association, LVEDD: left ventricular end-diastolic diameter, LVEF: left ventricular ejection fraction, FU: follow-up

Supplementary Table 5: Mixed-effects regression models for echocardiographic parameters and functional class over time

Parameter	Discharge	p	1-Year FU	p	2-Year FU	p
	Estimate (95% CI)		Estimate (95% CI)		Estimate (95% CI)	
MR	0.00015 (0.00003–0.00089)	<0.001	0.00017 (0.00003–0.00107)	<0.001	0.00014 (0.00002–0.00108)	<0.001
NYHA	–	–	0.02 (0.01–0.05)	<0.001	0.03 (0.01–0.11)	<0.001
LVEF	-0.30 (-0.42, -0.17)	<0.001	-0.31 (-0.47, -0.16)	<0.001	-0.18 (-0.39, 0.04)	0.112
MPG	+0.64 (0.45, 0.83)	<0.001	+0.94 (0.72, 1.17)	<0.001	+0.95 (0.63, 1.27)	<0.001
LVEDD	-0.31 (-0.47, -0.14)	<0.001	-0.24 (-0.44, -0.04)	0.020	-0.50 (-0.78, -0.22)	<0.001
LVEDV	-0.29 (-0.48, -0.10)	0.003	-0.23 (-0.45, -0.01)	0.038	-0.47 (-0.78, -0.16)	0.003
LVESD	-0.13 (-0.33, 0.06)	0.188	+0.05 (-0.18, 0.28)	0.668	-0.17 (-0.49, 0.14)	0.287
LVESV	-0.10 (-0.22, 0.02)	0.102	-0.03 (-0.18, 0.11)	0.659	-0.11 (-0.33, 0.10)	0.298
TAPSE	-0.04 (-0.29, 0.20)	0.736	-0.19 (-0.47, 0.09)	0.186	-0.12 (-0.51, 0.28)	0.555
PASP	-0.10 (-0.32, 0.12)	0.357	-0.33 (-0.59, -0.08)	0.009	-0.37 (-0.71, -0.03)	0.033

Baseline was the reference level in all models and NYHA was not measured at discharge. Estimates for CLMM (MR, NYHA) are OR and β for the remaining LMMs.

Abbreviations:

LVEDD: left ventricular end-diastolic diameter, LVEDV: left ventricular end-diastolic volume, LVEF: left ventricular ejection fraction, LVESD: left ventricular end-systolic diameter, LVESV: left ventricular end-systolic volume, MPG: mean transvalvular pressure gradient, MR: mitral regurgitation, NYHA: New York Heart Association, PASP: pulmonary artery systolic pressure, TAPSE: tricuspid annular plane systolic excursion

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Supplementary Table 6: Random effects from mixed-effects models for echocardiographic parameters and functional class over time

Parameter	σ^2	τ_{00}	ICC	Marginal R^2 / Conditional R^2	N
MR	3.29	1.67	0.34	0.78 / 0.85	124
NYHA	3.29	3.06	0.48	0.38 / 0.68	124
LVEF	113.35	85.70	0.43	0.02 / 0.80	119
MPG	2.58	1.08	0.30	0.21 / 0.38	123
LVEDD	26.52	39.62	0.60	0.03 / 0.67	121
LVESD	28.49	68.80	0.71	0.01 / 0.71	94
LVEDV	744.28	1266.57	0.63	0.03 / 0.66	91
LVESV	253.20	1510.18	0.86	0.01 / 0.81	92
TAPSE	12.87	11.62	0.47	0.01 / 0.56	102
PASP	113.35	85.70	0.43	0.02 / 0.48	119

Abbreviations:

LVEDD: left ventricular end-diastolic diameter, LVEDV: left ventricular end-diastolic volume, LVEF: left ventricular ejection fraction, LVESD: left ventricular end-systolic diameter, LVESV: left ventricular end-systolic volume, MPG: mean transvalvular pressure gradient, MR: mitral regurgitation, NYHA: New York Heart Association, PASP: pulmonary artery systolic pressure, TAPSE: tricuspid annular plane systolic excursion

Supplementary Table 7: Univariable and multivariable Cox regression analysis for all-cause mortality and HF rehospitalization at 2 years

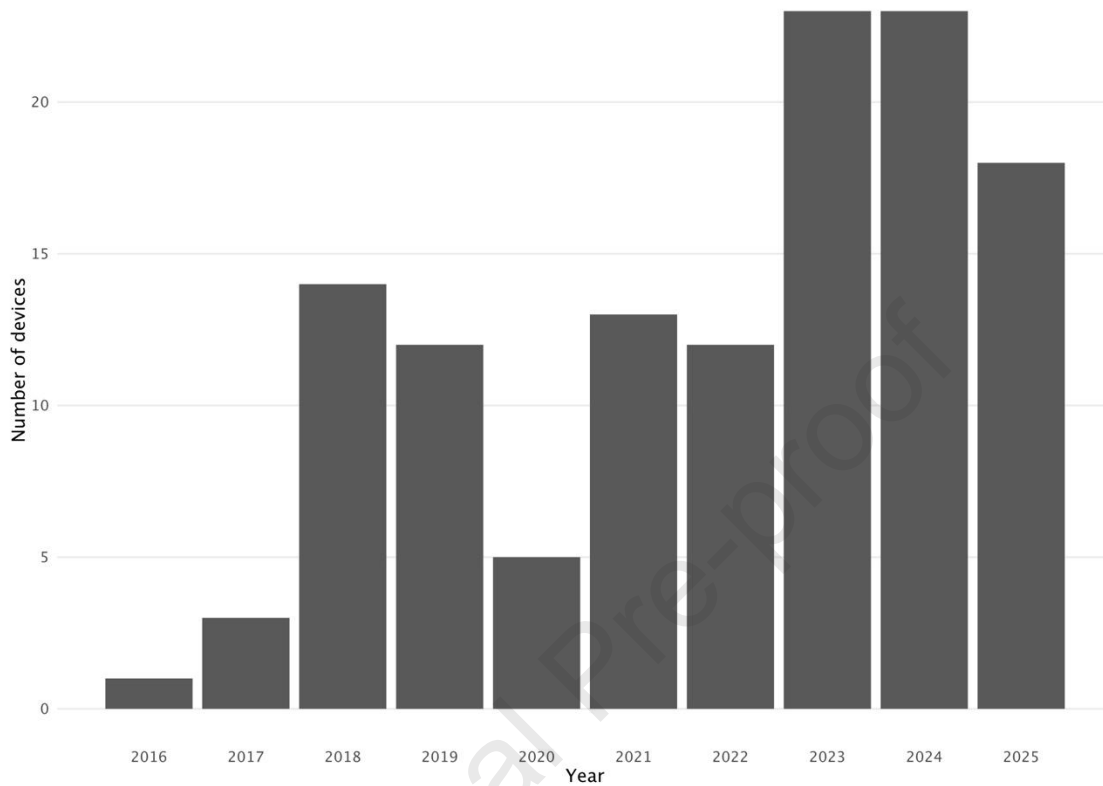
Parameters	Univariable (N=124)		Multivariable (N=108)	
	Hazard Ratio (95%-CI)	p-value	Hazard Ratio (95%-CI)	p-value
Age	0.99 (0.95, 1.03)	0.553		
Female sex	1.11 (0.63, 1.94)	0.721		
Body Mass Index, kg/m ²	1.07 (1.00, 1.14)	0.041	1.05 (0.99,1.12)	0.103
Arterial hypertension	0.99 (0.47, 2.12)	0.986		
Atrial fibrillation	4.67 (1.44, 15.11)	0.010	3.97 (1.22, 12.96)	0.022
Coronary artery disease	1.40 (0.73, 2.67)	0.308		
Extracardiac arteriopathy	0.66 (0.20, 2.16)	0.495		
COPD	0.93 (0.46, 1.86)	0.827		
Diabetes	0.95 (0.52, 1.74)	0.858		
Prior CABG	0.57 (0.29, 1.12)	0.106		
Prior PCI	0.77 (0.43, 1.38)	0.385		
Prior myocardial infarction	0.69 (0.37, 1.30)	0.252		
Prior stroke	1.10 (0.39, 3.05)	0.858		
Prior dialysis	0.00 (0.00, Inf)	0.996		
eGFR, ml/min/1.73m ²	0.98 (0.97, 1.0)	0.044	0.99 (0.97, 1.00)	0.078
NYHA functional class III/IV	1.10 (0.53, 2.27)	0.793		
MR etiology (secondary MR)	0.70 (0.38, 1.27)	0.235		
EROA, cm ²	0.92 (0.11, 7.62)	0.935		

Regurgitant volume, mL	1.01 (0.88, 1.02)	0.466
MPG, mmHg	1.01 (0.88, 1.16)	0.860
LVEF, %	1.01 (0.99, 1.03)	0.444
SVI, ml/m ²	0.99 (0.97, 1.01)	0.404
LVEDD, mm	0.98 (0.94, 1.01)	0.193
LVEDV, mL	0.99 (0.98, 1.00)	0.168
LVESD, mm	0.98 (0.95, 1.02)	0.385
LVESV, mL	0.99 (0.98, 1.00)	0.171
LAV, mL	1.01 (1.00, 1.02)	0.062
TAPSE, mm	1.01 (0.94, 1.09)	0.809
PASP, mmHg	0.98 (0.97, 1.00)	0.127
TR $\geq 2+$	0.92 (0.51, 1.67)	0.780

Left atrial volume (LAV) was excluded from the multivariable model because of more than 20% missing data.

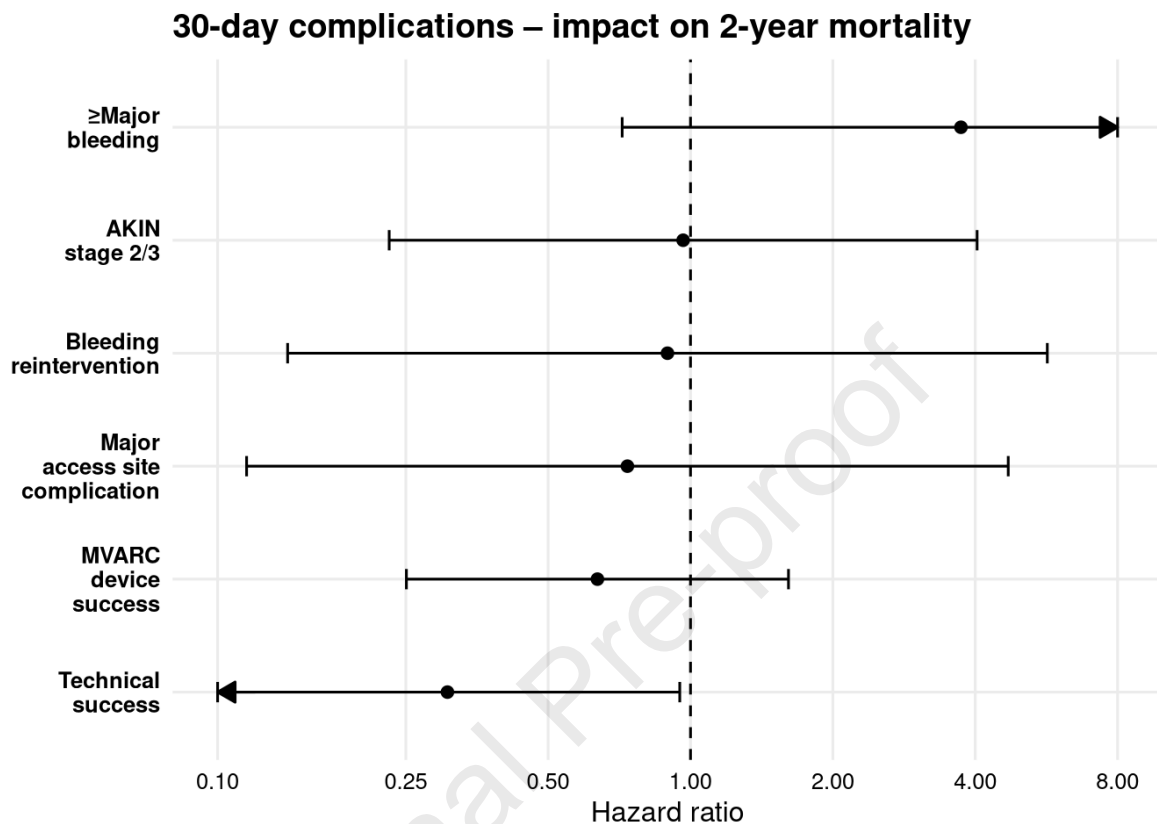
Abbreviations:

CABG: coronary artery bypass graft, COPD: chronic obstructive pulmonary disease, eGFR: estimated glomerular filtration rate, EROA: effective regurgitant orifice area, LAV: left atrial volume, LVEDD: left ventricular end-diastolic diameter, LVEDV: left ventricular end-diastolic volume, LVEF: left ventricular ejection fraction, LVESD: left ventricular end-systolic diameter, LVESV: left ventricular end-systolic volume, MPG: mean transvalvular pressure gradient, MR: mitral regurgitation, NYHA: New York Heart Association, PASP: pulmonary artery systolic pressure, PCI: percutaneous coronary intervention, SVI: stroke volume index, TAPSE: tricuspid annular plane systolic excursion, TR: tricuspid regurgitation

Supplementary Figure 1: Annual distribution of transfemoral TMVR devices.

Legend: Bar chart illustrating the number of transfemoral transcatheter mitral valve replacement (TF TMVR) devices implanted per year between 2016 and 2025. The use of TF devices increased over time, with a marked rise in procedural volume in the later years of the study period.

Supplementary Figure 2: Adjusted impact of 30-day complications on 2-year all-cause mortality



Legend: Cox regression analyses for 2-year-all-cause mortality were performed to evaluate the impact of procedural complications and 30-day MVARC-defined outcomes, adjusted for baseline predictors identified in the multivariable model. Adjusted hazard ratios with 95% confidence intervals are presented in the forest plot; truncated confidence intervals are indicated by arrows. Technical success was independently associated with improved 2-year survival.

Abbreviations: AKIN: Acute Kidney Injury Network, CAD: coronary artery disease; COPD: chronic obstructive pulmonary disease, eGFR: estimated glomerular filtration rate, LVOT: left ventricular outflow obstruction, MR: mitral regurgitation, MVARC: Mitral Valve Academic Research Consortium

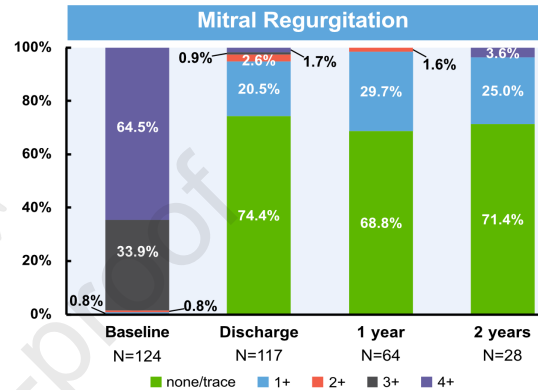
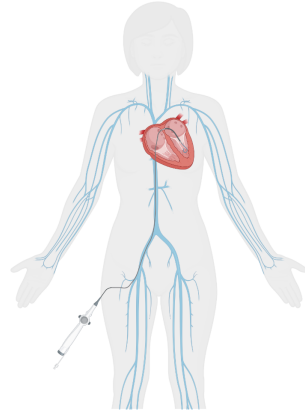
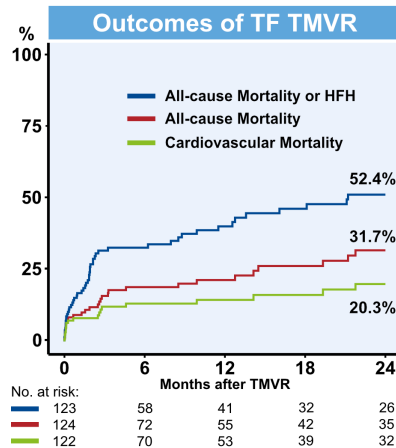
Clinical Outcomes of Transfemoral Transcatheter Mitral Valve Replacement

High-Risk Patients undergoing TF TMVR
Between October 2016 and December 2025
(N=124)

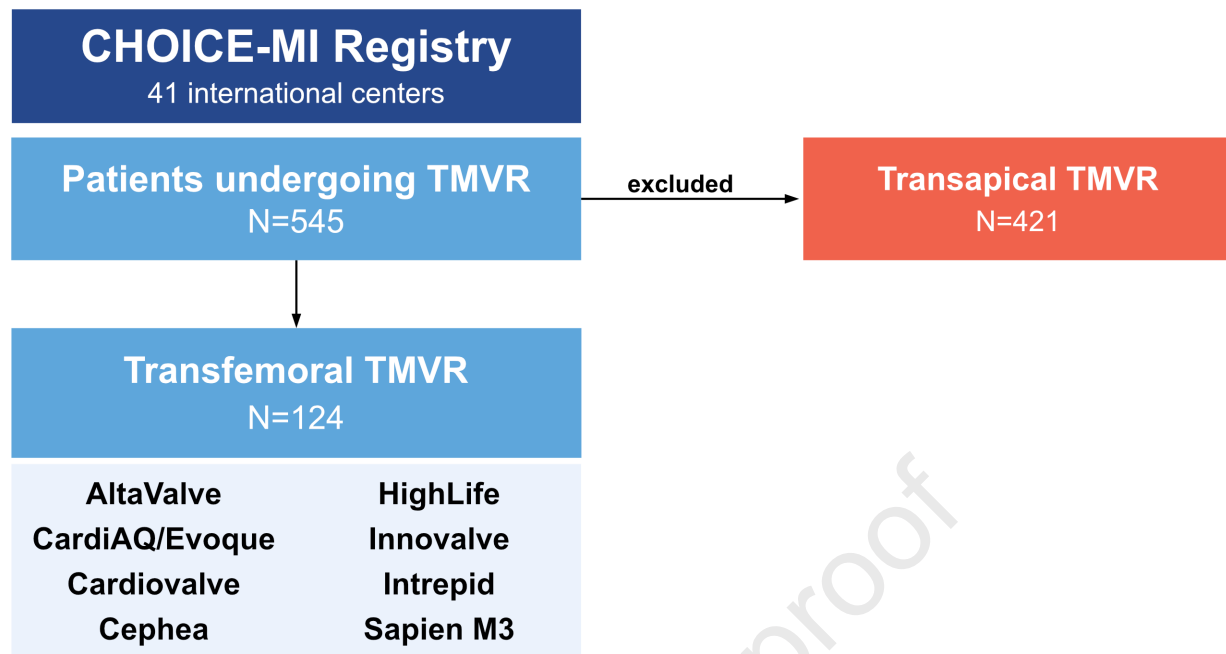
CHOICE-MI Registry

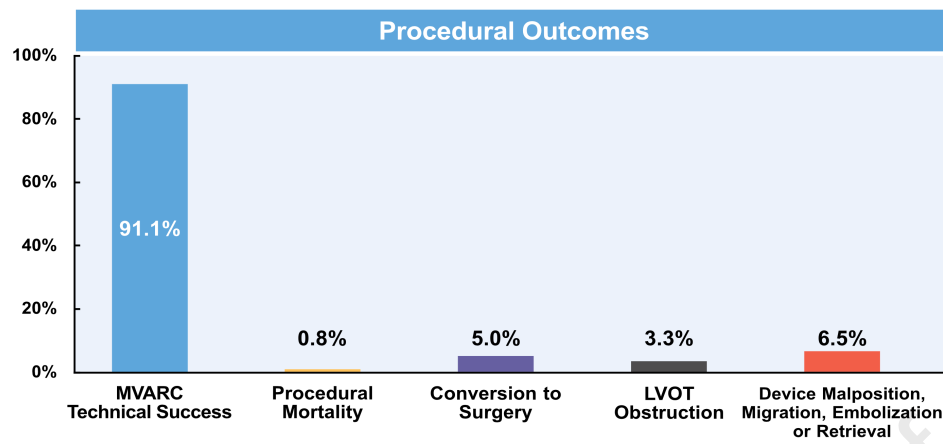
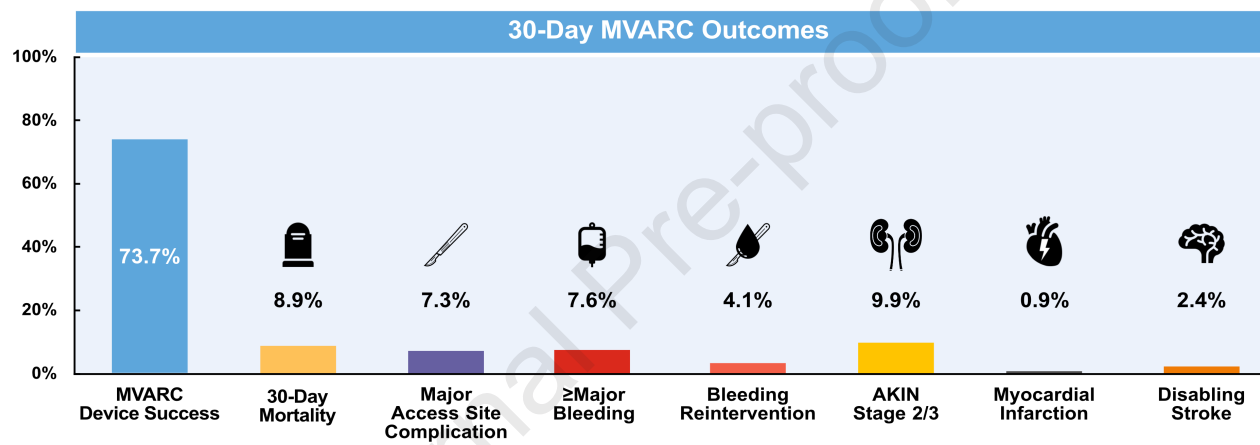
International, multicenter registry

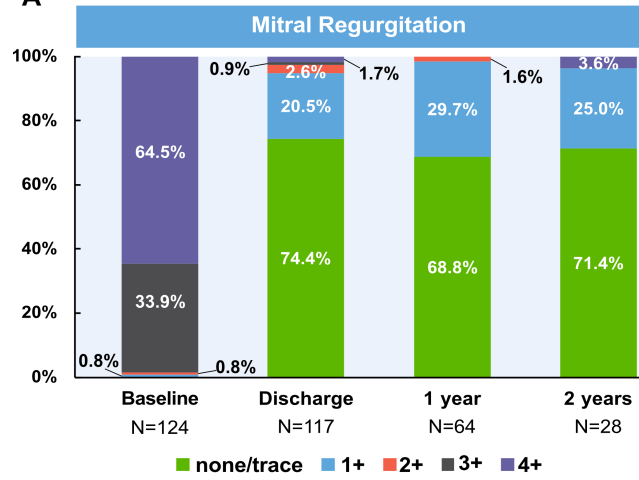
41 centers in Europe/UK,
North America, and Australia



- Transfemoral TMVR using 9 different dedicated devices was performed with high technical success (91.1%) and low procedural mortality (0.8%).
- MR reduction to $\leq 1+$ was achieved in approximately 95% of patients and remained durable through 2-year follow-up.
- Elimination of MR translated into left ventricular remodeling at follow-up.
- Technical success was associated with survival, whereas atrial fibrillation was an independent predictor of all-cause mortality and heart failure hospitalization.



A**B**

A**B**